Topic 13

Single-factor and factorial designs

Overview

- Single-factor designs
- Factorial designs
- Experimental research

Single-factor designs

- How to assign subjects to the different levels of the independent variable(s)?
  - Between-subjects design (independent-groups design)
    - Separate groups of subjects receive different levels of the independent variable(s).
  - Within-subjects design (within-groups /repeated-measures design)
    - All subjects receive all levels of the independent variable(s).

Within & Between Subjects Designs

Independent variable	**Dependent variable**
Location of lesion in the cortex	% correct
Rotation speed of radar arm	% detected
Learning method	Performance on test
Odor	Score concerning the attractiveness of person

Single-factor designs

- Between-subjects design

Independent variable

Level 1

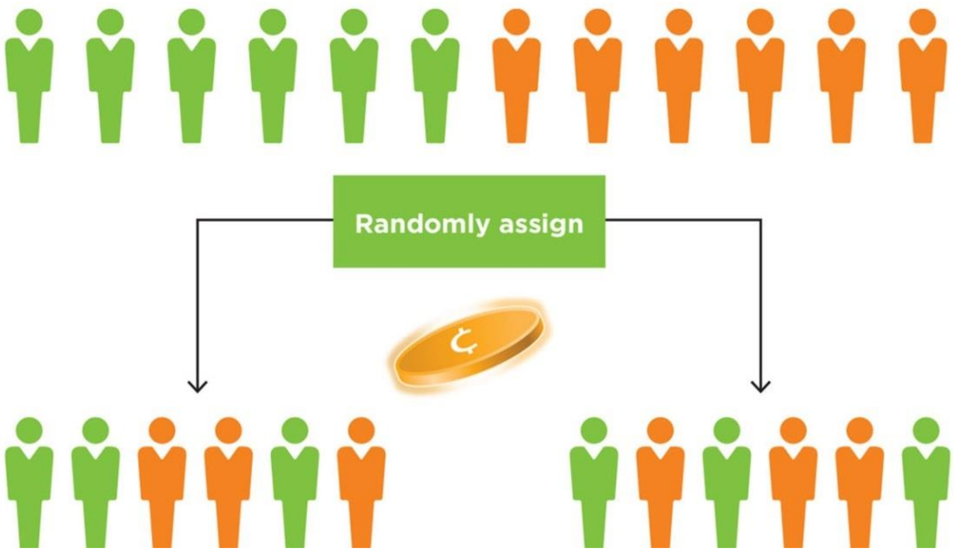
Level 2

Single-factor designs

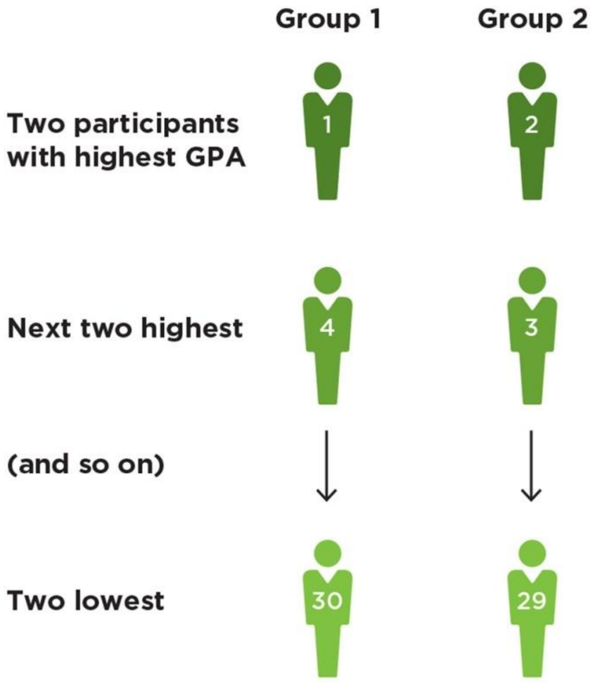
- Between-subjects design
  - Advantage
    - There is no chance that one level of the independent variable affects performance under the other level of the independent variable.
  - Disadvantage
    - There is a chance that groups differ.
      - Therefore
        - random assignment
        - matching

Single-factor designs

Random assignment



Matching



Single-factor designs

- Between-subjects design
  - Use when
    - the independent variable has a permanent effect.
    - there is a large risk of order effects
      - Practice
      - Carryover

Single-factor designs

- Within-subjects design

Independent variable

Level 1

Level 2

Single-factor designs

- Within-subjects design
  - Advantage
    - There is no chance of group differences:
      - each subject serves as their own control.
  - Disadvantage
    - There is a chance of order effects (practice, carryover)
      - Therefore
        - Complete counterbalanced design (full counterbalancing)
        - Latin Square design
        - Balanced Latin Square design
        - Random order

Partial counterbalancing

Single-factor designs

- Within-subjects design
  - Complete counterbalanced design
    - e.g., 4 levels

1 2 3 4	1 2 4 3	1 3 2 4	1 3 4 2
1 4 2 3	1 4 3 2	2 1 3 4	2 1 4 3
2 3 1 4	2 3 4 1	2 4 1 3	2 4 3 1
3 1 2 4	3 1 4 2	3 2 1 4	3 2 4 1
3 4 1 2	3 4 2 1	4 1 2 3	4 1 3 2
4 2 1 3	4 2 3 1	4 3 1 2	4 3 2 1

Single-factor designs

- Within-subjects design
  - Latin-square design

    1 2 3 4

    2 3 4 1

    3 4 1 2

    4 1 2 3
  - Balanced Latin-square design

    1 2 3 4

    2 4 1 3

    3 1 4 2

    4 3 2 1

Single-factor designs

- Within-subjects design
  - Random order

    4 8 1 3 9 5 2 6 7 10
    7 10 1 9 6 5 4 8 2 3
    5 9 2 4 8 7 3 10 1 6 etc.

Single-factor designs

- Within-subjects design
  - Use when
    - the independent variable does not have a permanent effect.
    - order effects (practice, carryover) can be avoided by counterbalancing or random order.
- Within-subject designs are more efficient than between-subjects designs (more power).

Single-factor designs

A scientist performs an experiment to investigate the influence of display background colors on the speed with which people can read white-colored texts.

17 different background colors are examined in a within-subjects design with 9 subjects.

The following are <u>not used</u>

(a) a complete counterbalanced design

(b) a Latin-square design

Why not?

In which orders should he present the different levels of the independent variable to the individual subjects?

Factorial designs

- Design in which each level of every independent variable occurs with all levels of the other independent variable
  - Example
    - Blood Alcohol Content (BAC) with three levels:
      - 0‰, 0.5‰, 1.0‰ (per mil)
    - Power steering with two levels:
      - with/without

Factorial designs

Power Steering	Blood Alcohol Content		
	0	0.5	1.0
without	without, 0	without, 0.5	without, 1.0
with	with, 0	with, 0.5	with, 1.0

- Factorial combination of all levels of BAC (3) and Power Steering (2)
- This results in $3 \times 2 = 6$ conditions (number of cells in table).
- BAC and Power Steering vary independently of each other.
- They are not confounded.

Factorial designs

	A1	A2	A3
B1	A1B1	A2B1	A3B1
B2	A1B2	A2B2	A3B2

General format

- Letters: factors (independent variables) – A, B, C, ...
- Figures: level of factor – 1, 2, 3, ...
- Example of a "3 × 2 factorial design"

Factor A has 3 levels

Factor B has 2 levels

- There are 3 x 2 = 6 conditions

Factorial designs

C1

	A1	A2
B1	A1B1C1	A2B1C1
B2	A1B2C1	A2B2C1

C2

	A1	A2
B1	A1B1C2	A2B1C2
B2	A1B2C2	A2B2C2

General format

- Three factors (A, B, C), all with two levels (1, 2)
- $2 \times 2 \times 2$ factorial design
- Eight conditions

Example Factor C: Road complexity (10 vs 20 cones)

Factorial designs

	A1	A2	A3
B1	A1B1	A2B1	A3B1
B2	A1B2	A2B2	A3B2

Different designs

- **Between-subjects design**: A and B are both between-subjects factors. Six groups.
- **Within-subjects design**: A and B are both within-subjects factors, e.g., six sequences (in Latin Square).
- **Mixed design**: One of the factors is a between-subjects factor; the other factor is a within-subjects factor.

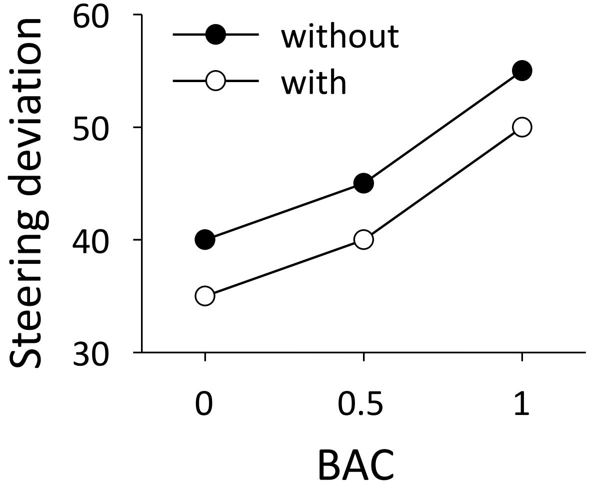
Factorial designs

	A1	A2	A3
B1	A1B1	A2B1	A3B1
B2	A1B2	A2B2	A3B2

Advantages Factorial design

- **Efficiency**
  - e.g., one experiment with two factors instead of two experiments, each with one factor.
- **Generalization**
  - e.g., the effect of BAC occurs both in cars with and without power steering.
- **Information on interaction between factors**
  - e.g., does the effect of BAC depend on the presence of power steering?

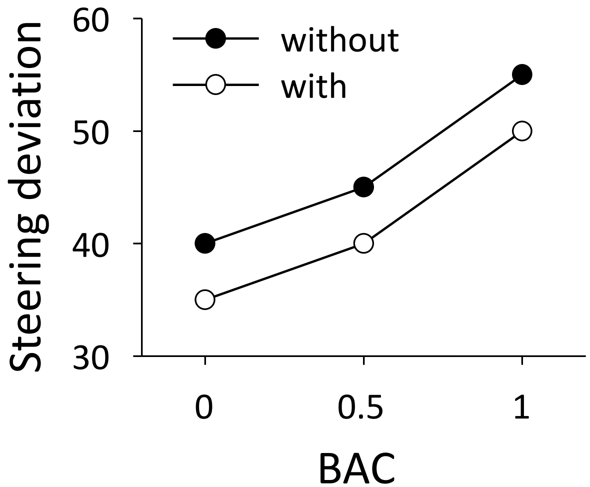
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	40	50

- Steering deviation (y-axis) as a function of the factors:
  - BAC (x-axis)
  - Presence of power steering (→ legend)
  - Table contains the same information

Factorial designs

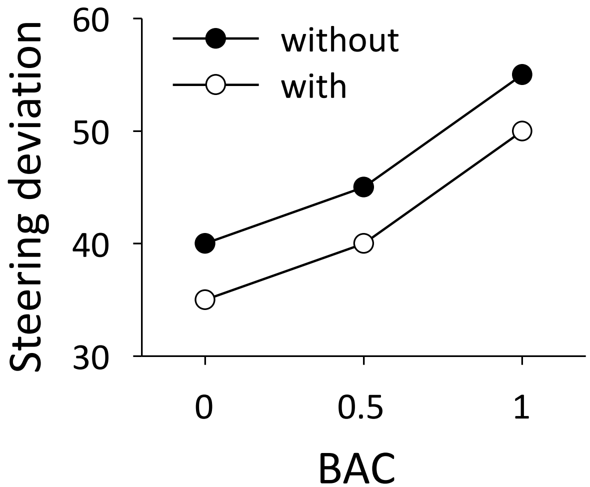


	0	0.5	1.0
without	40	45	55
with	35	40	50

Effects in a two-factorial design

- Main effect of Factor A (BAC)
- Main effect of Factor B (Power steering)
- Interaction effect A $\times$ B

Factorial designs

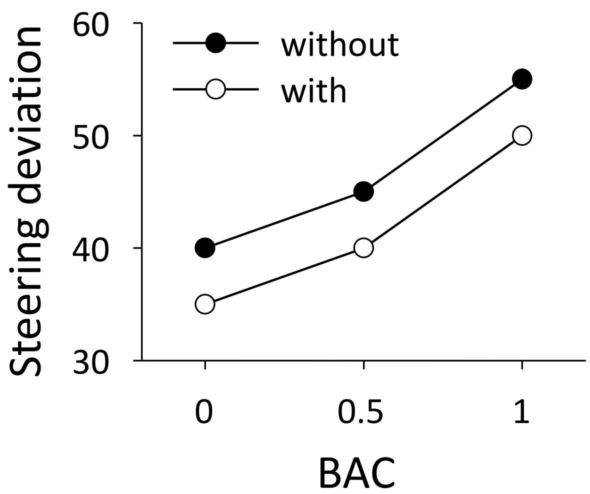


	0	0.5	1.0
without	40	45	55
with	35	40	50

Main effect

The overall effect of one independent variable on the dependent variable, averaged over the levels of the other independent variable

Factorial designs

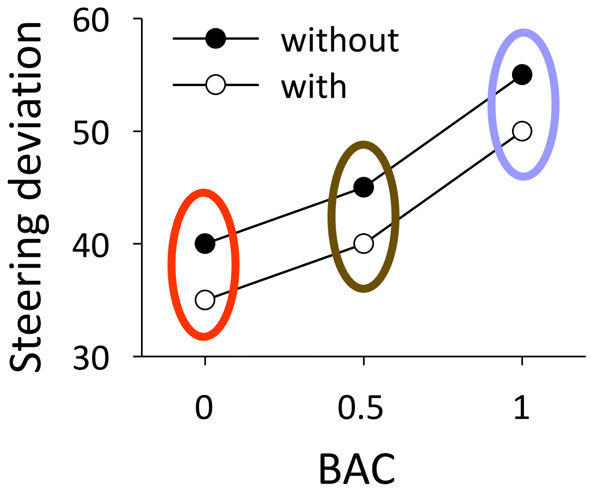


	0	0.5	1.0
without	40	45	55
with	35	40	50

Is there a main effect of BAC?

- Calculate the average over the different levels of power steering

Factorial designs

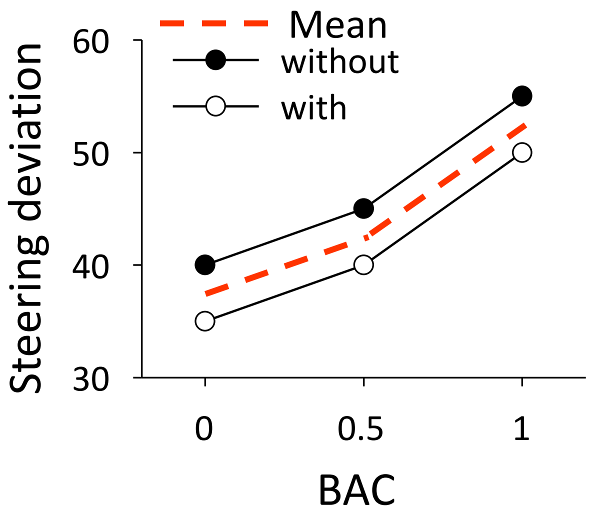


	0	0.5	1.0
without	40	45	55
with	35	40	50

Is there a main effect of BAC?

- Calculate the average over the different levels of power steering

Factorial designs

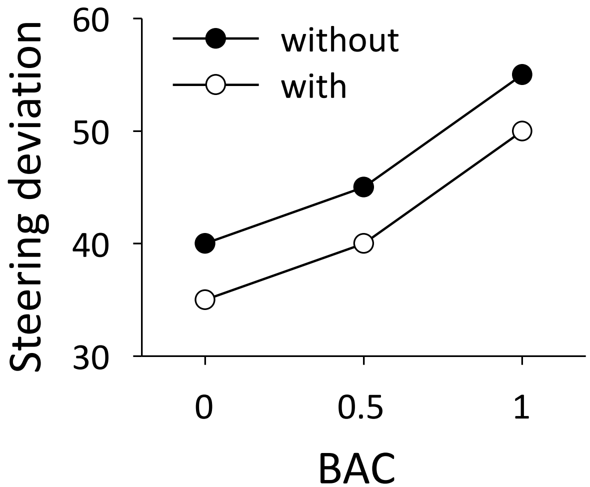


	0	0.5	1.0
without	40	45	55
with	35	40	50
Mean	37.5	42.5	52.5

Is there a main effect of BAC?

- Calculate the average over the different levels of power steering
  → Main effect of BAC

Factorial designs

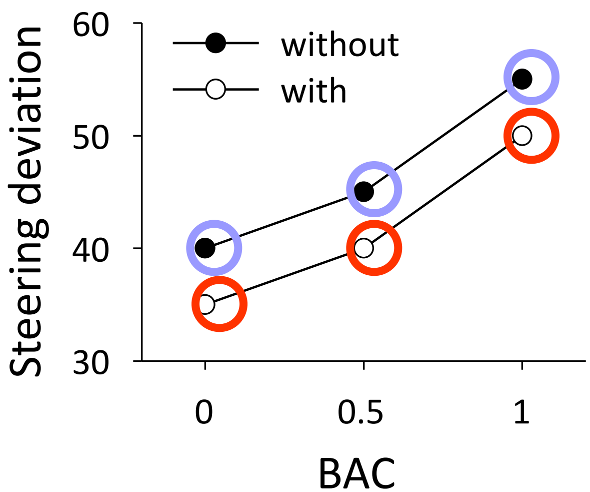


	0	0.5	1.0
without	40	45	55
with	35	40	50

Is there a main effect of Power steering?

- Calculate the average over the different levels of BAC

Factorial designs

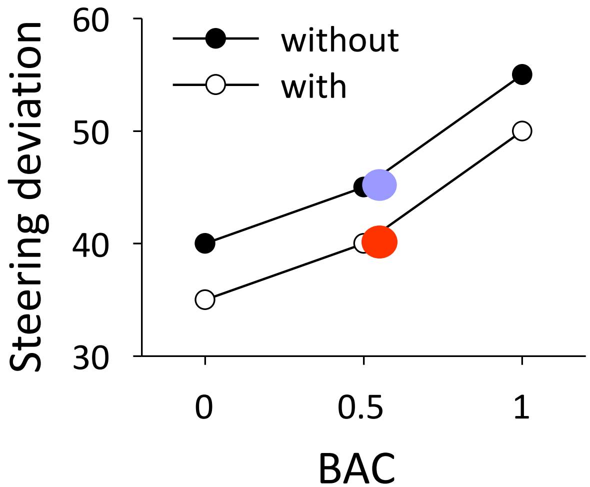


	0	0.5	1.0
without	40	45	55
with	35	40	50

Is there a main effect of Power steering?

- Calculate the average over the different levels of BAC

Factorial designs

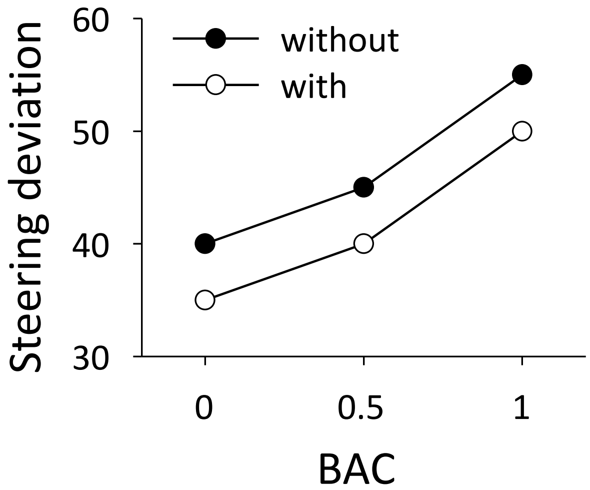


	0	0.5	1.0	mean
without	40	45	55	47
with	35	40	50	42

Is there a main effect of Power steering?

- Calculate the average over the different levels of BAC
  → Main effect of Power steering

Factorial designs



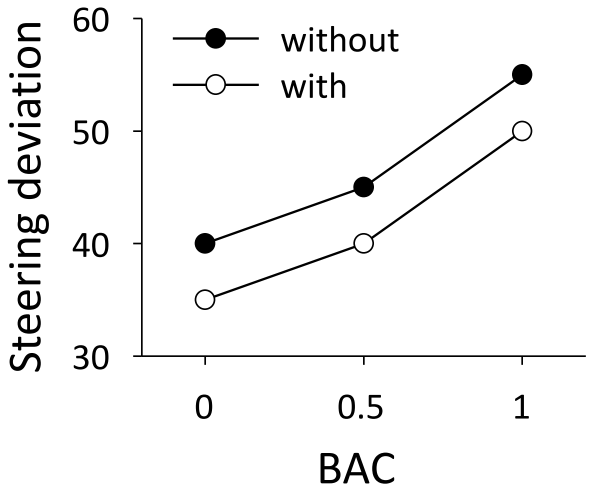
	0	0.5	1.0
without	40	45	55
with	35	40	50

Interaction effect

When the effect of one independent variable depends on the level of the other independent variable(s).

- Symmetrical: interaction A x B = Interaction B x A

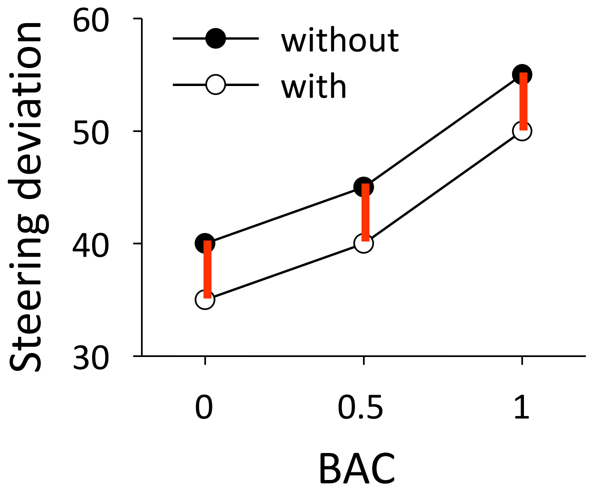
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	40	50

Is there an interaction BAC x Power steering?

Factorial designs



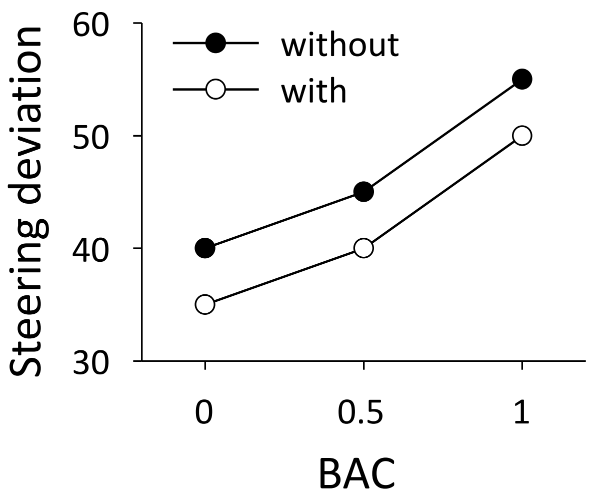
	0	0.5	1.0
without	40	45	55
with	35	40	50
Difference	5	5	5

Is there an interaction BAC x Power steering?

The effect of Power steering does not depend on BAC → no interaction

- Graph: parallel lines (with vs. without)

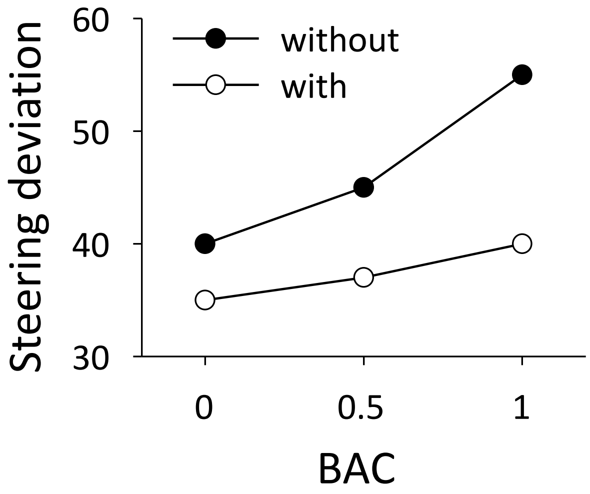
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	40	50

- **Main effect of BAC** ✓
- **Main effect of Power steering** ✓
- **Interaction BAC x Power steering** ✗

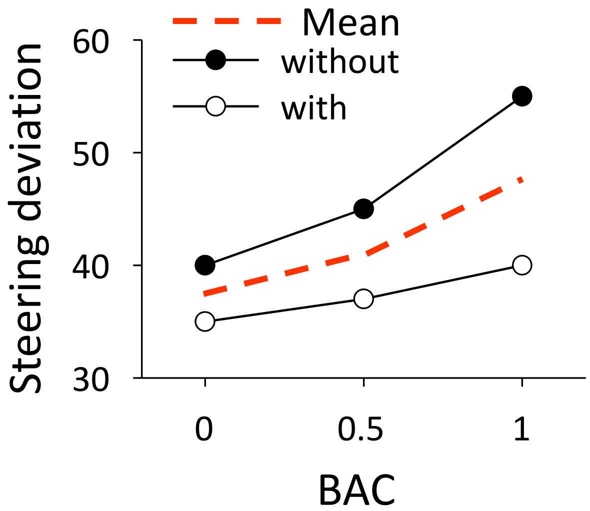
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	37	40

- **Main effect of BAC?**
- Main effect of Power steering?
- Interaction BAC x Power steering?

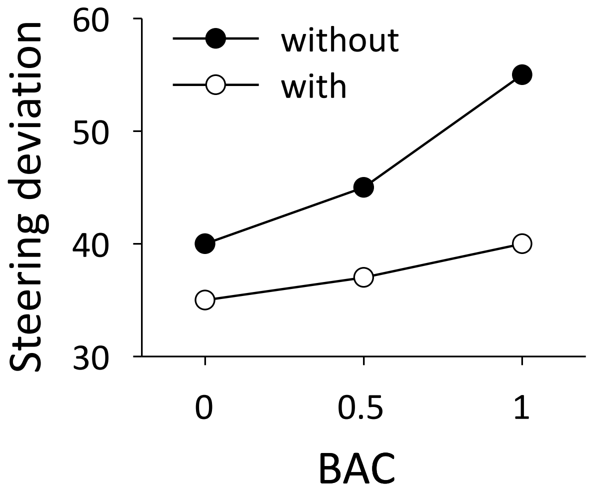
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	37	40
Mean	37.5	41	47.5

- **Main effect of BAC?** ✔
- Main effect of Power steering?
- Interaction BAC x Power steering?

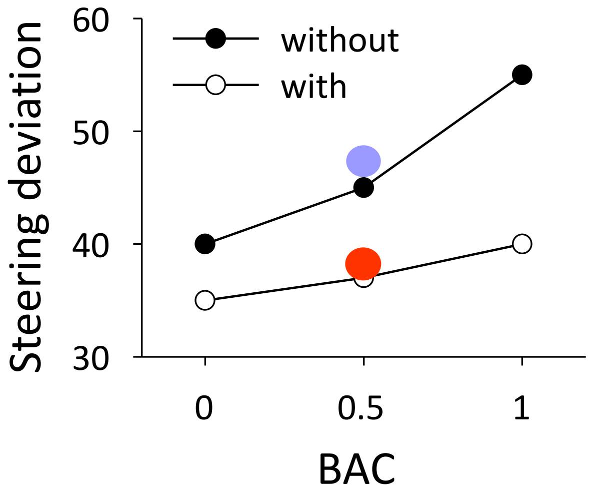
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	37	40

- Main effect of BAC? ✔
- **Main effect of Power steering?**
- Interaction BAC x Power steering?

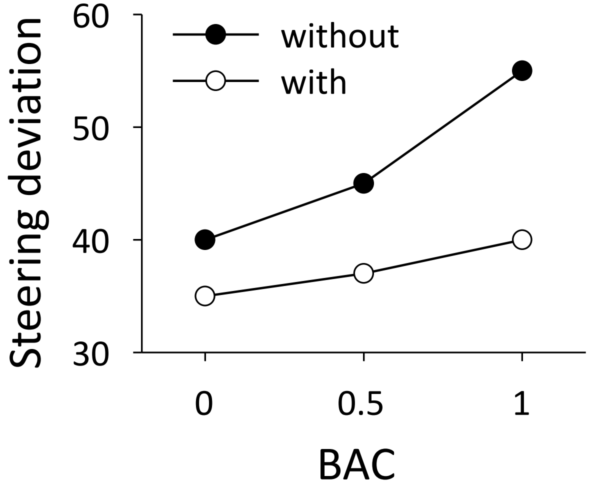
Factorial designs



	0	0.5	1.0	mean
without	40	45	55	47
with	35	37	40	38

- Main effect of BAC?
- **Main effect of Power steering?**
- Interaction BAC x Power steering?

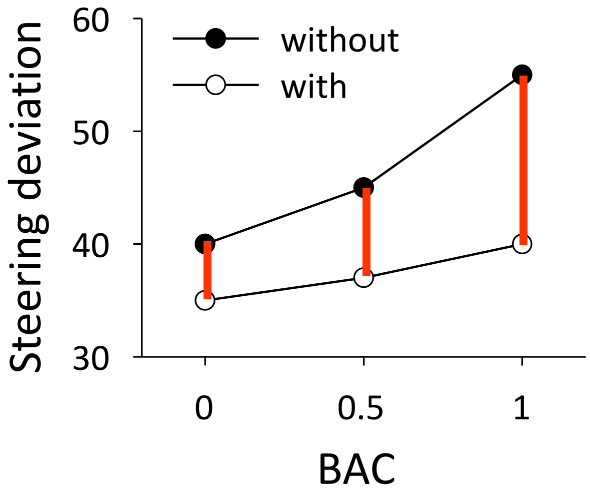
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	37	40

- Main effect of BAC? ✔
- Main effect of Power steering? ✔
- **Interaction BAC x Power steering?**

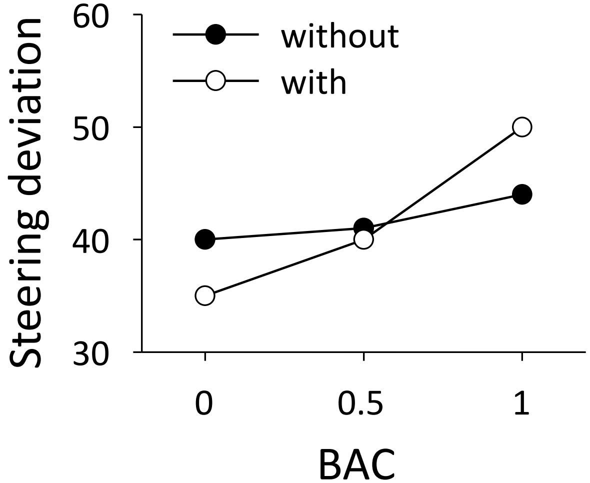
Factorial designs



	0	0.5	1.0
without	40	45	55
with	35	37	40
Difference	5	8	15

- Main effect of BAC? ✓
- Main effect of Power steering? ✓
- **Interaction BAC x Power steering?** ✓

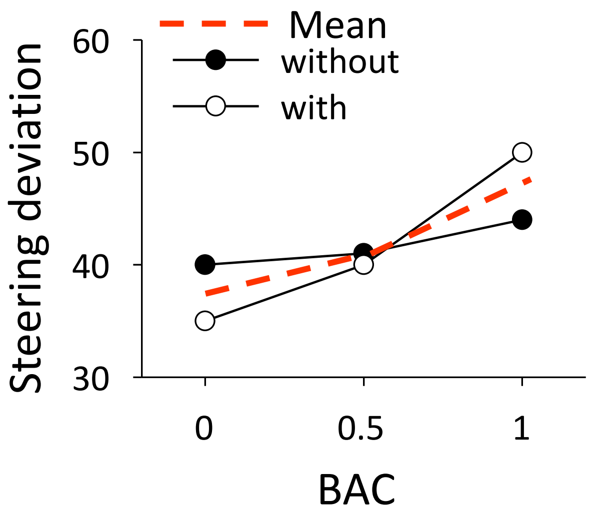
Factorial designs



	0	0.5	1.0
without	40	41	44
with	35	40	50

- **Main effect of BAC?**
- Main effect of Power steering?
- Interaction BAC x Power steering?

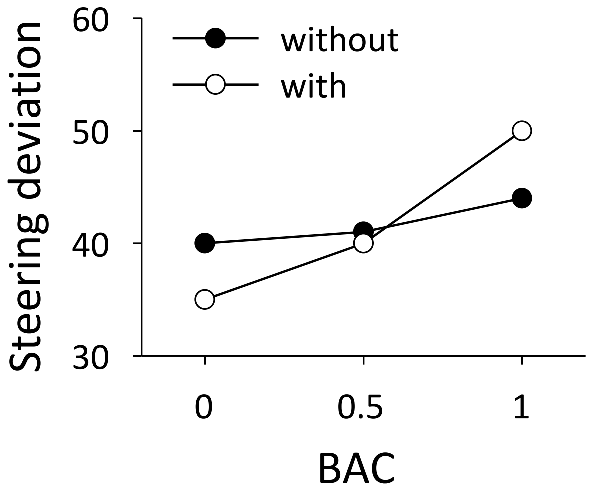
Factorial designs



	0	0.5	1.0
without	40	41	44
with	35	40	50
Mean	37.5	40.5	47

- **Main effect of BAC?** ✔
- Main effect of Power steering?
- Interaction BAC x Power steering?

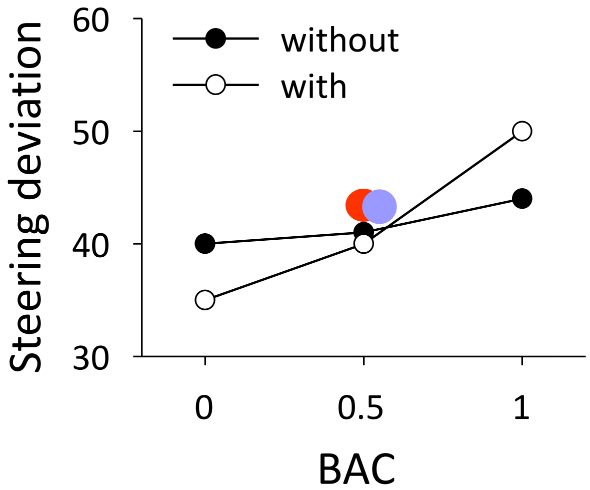
Factorial designs



	0	0.5	1.0
without	40	41	44
with	35	40	50

- Main effect of BAC? ✔
- **Main effect of Power steering?**
- Interaction BAC x Power steering?

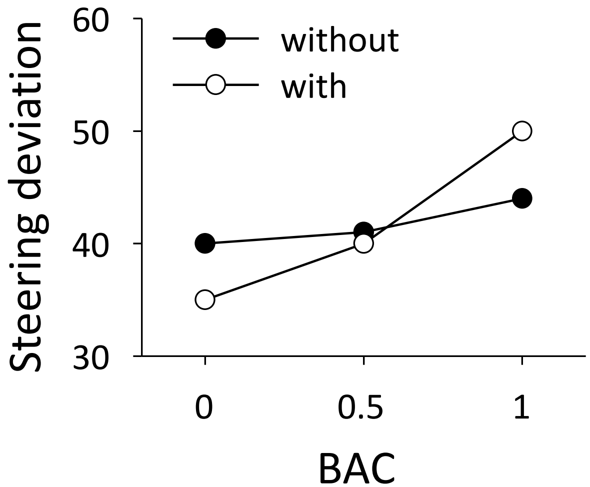
Factorial designs



	0	0.5	1.0	mean
without	40	41	44	42
with	35	40	50	42

- Main effect of BAC? ✔
- **Main effect of Power steering?** ✗
- Interaction BAC x Power steering?

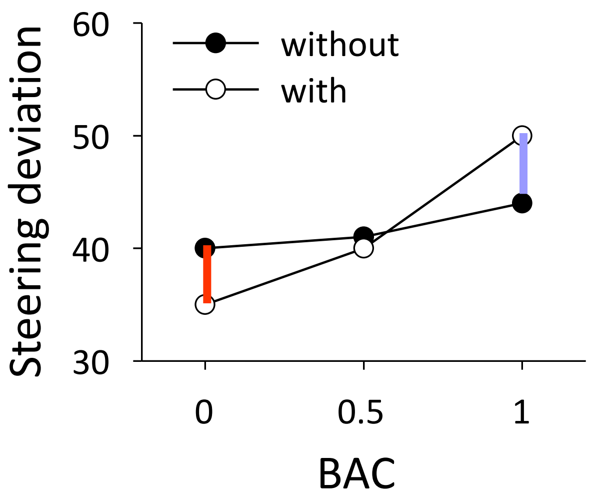
Factorial designs



	0	0.5	1.0
without	40	41	44
with	35	40	50

- Main effect of BAC? ✔
- Main effect of Power steering? ✘
- **Interaction BAC x Power steering?**

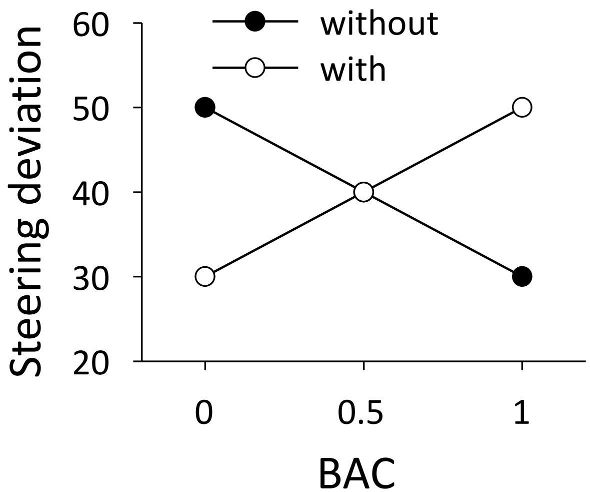
Factorial designs



	0	0.5	1.0
without	40	41	44
with	35	40	50
Difference	5	1	-6

- Main effect of BAC? ✓
- Main effect of Power steering? ✗
- **Interaction BAC x Power steering?** ✓

Factorial designs



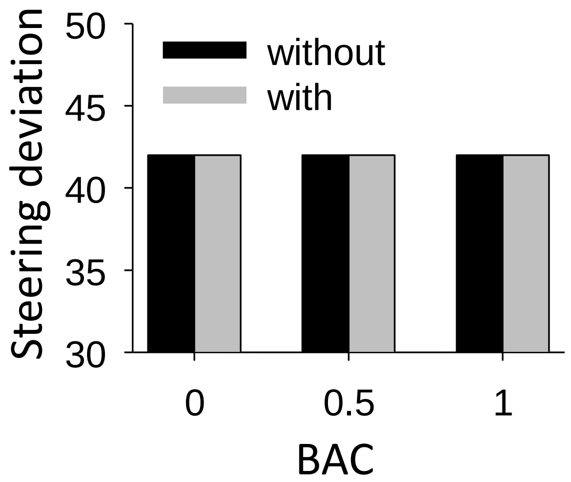
- **Main effect of BAC?** ✗
- **Main effect of Power steering?** ✗
- **Interaction BAC x Power steering?** ✓

Factorial designs

		BAC		
		0	0.5	1.0
Power steering	without	65	60	50
	with	35	40	50

- **Main effect of BAC?** ✗
- **Main effect of Power steering?** ✓
- **Interaction BAC x Power steering?** ✓

Factorial designs



- **Main effect of BAC?** ✕
- **Main effect of Power steering?** ✕
- **Interaction BAC x Power steering?** ✕

Factorial designs

- Which effects can be distinguished in a factorial design with three independent variables A, B, C?
    - Main effect A
    - Main effect B
    - Main effect C
    - Interaction A x B
    - Interaction A x C
    - Interaction B x C
    - Interaction A x B x C (three-way interaction)

Experimental research

- Experimental research → causal claims
  - Experimental research establishes internal validity
  - However, this requires appropriate designs!
    - Were there any design confounds?
      - **Between-subjects variables:**
        - Did the researcher control for selection effects using random assignment or matching?
      - **Within-subjects variables:**
        - Did the researcher control for order effects by (partial) counterbalancing or randomization?

Experimental research

- Experimental research → causal claims
  - Experimental research establishes internal validity
  - However, this requires appropriate designs!
  - It is also important to question how well
    - each variable was manipulated and measured (construct validity)
    - the results generalize (external validity)
    - the numbers support the conclusions (statistical validity)

Me after attending a 1 hour lecture